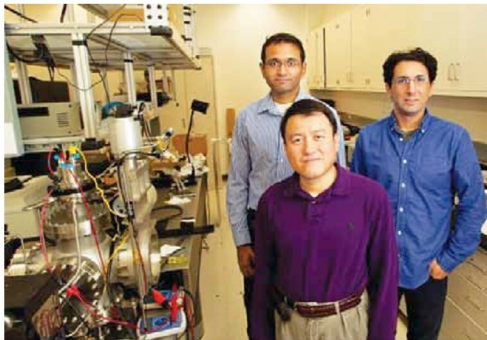




Professor Shanhui Fan (centre) with graduate students Aaswath Raman (left) and Eden Rephaeli (right). (Photo: Norbert von der Groeben)

bers. Several of these cells are lined up in stacks.

Recyclable solar cells developed from trees

We have reported about a lot of innovations concerning solar cells but this one will surely take you by surprise. Georgia Institute of Technology and Purdue University researchers have developed efficient solar cells using natural substrates derived from trees. What makes the news even more exciting is that the researchers have managed to recycle them in water by fabricating them on cellulose nanocrystal (CNC) substrates.

To date, organic solar cells have been typically fabricated on glass or plastic. Neither is easily recyclable, and petroleum-based substrates are not very eco-friendly.

The researchers report that the organic solar cells reach a power conversion efficiency of 2.7 per cent, an unprecedented figure for cells on substrates derived from renewable raw materials.

Key-chain sized charger

For all those who struggle with their cellphone's battery life, here's a good news. A UK-based company is working on a device that it touts as the

world's smallest emergency phone charging device. The key-chain sized device is just 3.3 cm (1.3-inch) in size, which makes it perfectly handy to be carried anywhere.

According to a Gizmag report, Devotec Industries Micro Fuel charger comes packed with a standard micro-USB port. It means that most of the phones can be charged with this ultra-portable charger. The charger is made up of die-cast aluminium, which makes it strong and light weight.

It is made up of an energy dense lithium-ion rechargeable battery, which is capable of providing around 20-30 minutes extra talk time, or a few hours on standby.

A solar structure to cool buildings

A new type of solar structure could vastly improve daylight cooling of buildings, cars and other structures by radiating sunlight back into space. With the growing need to look for energy conserving alternatives, researchers have developed a solar structure that is capable of improving daylight cooling of buildings, cars and other devices.

The design of the solution comprises a reflector, which enables optimum reflection of the sunlight.

According to a Science World report, the thermal emitter and solar reflector are combined into one device, which result in high performance rate of the system.

After testing, the researchers found that the new device is actually capable of achieving a net cooling power in excess of 100 watts per square metre.

3D mobile display that works without glasses

With the advent of 3D technology in television, smartphone and tablets, the entire viewing experience has been given a total makeover.

The basic trend of watching 3D content has been possible using 3D-enabled glasses, but companies are already looking at ways that would negate the use of such glasses, which can be cumbersome at times. Hewlett Packard (HP) has apparently taken a step in the 3D direction by developing a 3D screen for mobile devices that requires no glasses.

The viewer need not be directly in front and centre of the 3D TV in order to see the effect. HP's 3D display has been created in such a way that users can feel the 3D effect even at 45 degree viewing angle.

New coding language 'Rust' in development

Samsung has reportedly joined hands with Mozilla in an attempt to help the company develop a new programming language, Rust, which aims to replace the current 'Unsafe C++'.

Samsung has committed some of their technical know-how to Mozilla, once regarded as idealistic. They attempt to develop a new form of code, which will be able to handle today's heterogeneous, multicore hardware more effectively whilst also being more secure.

According to Mozilla, Rust will be equipped with relatively easy-to-use programming language primitives, and will be the simplest level of programming language expressions available to programmers. This will all presumably make it much easier for